← Back **Spectral Density Functions**
Graded Assignment • 3h

Due Jul 21, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Calculate the Fourier transform of function $X(t) = 2; -1 \leq t \leq 1$

1 / 1 point

- [ ] $\frac{2}{\pi\omega} e^{-\omega}$
- [ ] $\frac{2}{\pi\omega} \cos(\omega)$
- [x] $\frac{2}{\pi\omega} \sin(\omega)$

Correct
Correct! See below for details of the derivation.

$\hat{X}(\omega) = \frac{1}{2\pi} \int_{-1}^{1} 2e^{-i\omega t} dt = \frac{-1}{i\pi\omega}(e^{-i\omega} - e^{i\omega}) = \frac{2}{\pi\omega} \sin(\omega)$

- [ ] $\frac{2}{\pi\omega}$

2. In the code for estimating the spectral density function, we assume a spectral density function of the form: $S(\omega) = \frac{b}{2} e^{-b|\omega|}$

1 / 1 point

What is the covariance function $B(\tau)$ corresponding to this spectral density function?

- [ ] $B(\tau) = be^{-b\tau}$
- [ ] $B(\tau) = \sin(b\tau)$
- [x] $B(\tau) = \frac{b^2}{b^2+\tau^2}$

Correct
This is a bit tricky of a derivation (see below).$B(\tau) = \int_{-\infty}^{\infty} \frac{b}{2} e^{-b|\omega|} e^{i\omega\tau} d\omega = \frac{b}{2} \int_{-\infty}^{0} e^{\omega(b+i\tau)} d\omega + \frac{b}{2} \int_{0}^{\infty} e^{-\omega(b-i\tau)} d\omega = \frac{be^{\omega(b+i\tau)}}{2(b+i\tau)} |_{-\infty}^{0} + \frac{be^{-\omega(b-i\tau)}}{-2(b-i\tau)} |_{0}^{\infty} = \frac{b}{2(b+i\tau)} + \frac{b}{2(b-i\tau)} = \frac{b(b-i\tau)}{2(b^2+\tau^2)} + \frac{b(b+i\tau)}{2(b^2-\tau^2)} = \frac{b^2}{b^2+\tau^2}$

3. In the code for estimating the spectral density function, we assume a spectral density function of the form: $S(\omega) = \frac{b}{2} e^{-b|\omega|}$

1 / 1 point

What is the variance corresponding to this spectral density function?

- [x] 1

Correct
Correct! The variance is the integral of the spectral density function.

$\sigma^2 = \int_{-\infty}^{\infty} S(\omega) d\omega = b \int_{0}^{\infty} e^{-b\omega} d\omega = -e^{-b\omega}|_{0}^{\infty} = 0 - (-1) = 1$

- [ ] 0.5
- [ ] b

4. Question prompt goes here

1 / 1 point

In the code for estimating the spectral density function, we assume a spectral density function of the form: $S(\omega) = \frac{b}{2} e^{-b|\omega|}$

What is the correlation length corresponding to this spectral density function?

- ( ) 1
- ( ) b
- (•) πb
- ( ) D: b/2

Correct
Correct! The correlation length is the spectral density function evaluated at $\omega = 0$, multiplied by 2π, divided by the variance. We already found the variance is 1, so the correlation length is $2\pi b/2$, or πb.

5. In our code for estimating the spectral density function and the bandwidth, we averaged the estimates over 8 sample processes. If we add more sample processes, how do you expect the results to change?

1 / 1 point

- (•) There will be a decreased error.
- ( ) Error will stay the same.
- ( ) There will be an increased error.

Correct
Correct! The more samples we have, then the more data we have. Therefore, our resulting estimates should improve and the error should decrease.